THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 10 — MATHS · HARD LEVEL CHAPTER TEST SET 2

Chapter 10: **Circles** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. Two concentric circles have radii 70 cm and 42 cm. A chord of the larger circle touches the smaller circle. The length of the chord is: 1

(a) 112 cm **(b)** 28 cm **(c)** 84 cm **(d)** 56 cm

2. In $\triangle ABC$, right-angled at B, $AB = 30$ cm and $BC = 72$ cm. A circle is inscribed in the triangle. The radius of the inscribed circle is: 1

(a) 13 cm **(b)** 39 cm **(c)** 24 cm **(d)** 12 cm

3. A point P is 85 cm from the centre O of a circle of radius 75 cm. The length of the tangent drawn from P to the circle is: 1

(a) 1600 cm **(b)** 40 cm **(c)** 160 cm **(d)** 10 cm

4. The radius of the incircle of a right triangle with legs a, b and hypotenuse c is: 1

(a) $\dfrac{a+b}{2}$ **(b)** $\dfrac{a+b-c}{2}$ **(c)** $\dfrac{ab}{c}$ **(d)** $\dfrac{a+b+c}{2}$

5. Tangents PA and PB are drawn from a point P, 40 cm from the centre O of a circle of radius 24 cm. The length of each tangent is 32 cm. The length of the chord AB is: 1

(a) 48 cm **(b)** 19.2 cm **(c)** 64 cm **(d)** 38.4 cm

6. PA and PB are tangents from P to a circle, each of length 13 cm. A third tangent touches the circle at E on the minor arc AB and meets PA at C and PB at D. The perimeter of $\triangle PCD$ is: 1

(a) 39 cm **(b)** 15 cm **(c)** 26 cm **(d)** 13 cm

7. Tangents QA and QB are drawn from an external point Q to a circle, and $QA = (17x + 3)$ cm and $QB = (9x + 27)$ cm. The length of QA is: 1

(a) 78 cm **(b)** 3 cm **(c)** 56 cm **(d)** 54 cm

8. Two concentric circles have radii 73 cm and 48 cm. A chord of the larger circle touches the smaller circle. The length of the chord is: 1

(a) 50 cm **(b)** 121 cm **(c)** 110 cm **(d)** 55 cm

9. **Assertion (A):** The number of tangents that can be drawn to a circle from a point lying inside the circle is 2. 1

Reason (R): There is exactly one tangent to a circle at each point of the circle.

Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.

(b) Both A and R are true but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

10. **Assertion (A):** A line which touches a circle at exactly one point is a tangent to the circle. 1
Reason (R): A circle has infinitely many tangents.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. PA and PB are tangents from P to a circle, each of length 16 cm. A third tangent touches the circle at E on the minor arc AB and meets PA at C and PB at D. Find the perimeter of $\triangle PCD$. 2

12. Two concentric circles have radii 91 cm and 84 cm. A chord of the larger circle touches the smaller circle. Find the length of the chord. 2

13. Tangents PA and PB are drawn from a point P, 45 cm from the centre O of a circle of radius 36 cm. The length of each tangent is 27 cm. Find the length of the chord AB. 2

14. PA and PB are tangents from P to a circle, each of length 21 cm. A third tangent touches the circle at E on the minor arc AB and meets PA at C and PB at D. Find the perimeter of $\triangle PCD$. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Tangents PA and PB are drawn from a point P, 40 cm from the centre O of a circle of radius 24 cm. The length of each tangent is 32 cm. Find the length of the chord AB. 3

16. In $\triangle ABC$, right-angled at B, $AB = 51$ cm and $BC = 68$ cm. A circle is inscribed in the triangle. Find the radius of the inscribed circle. 3

17. Tangents QA and QB are drawn from an external point Q to a circle, and $QA = (16x + 9)$ cm and $QB = (9x + 30)$ cm. Find the length of QA. 3

OR

Tangents PA and PB are drawn from a point P, 75 cm from the centre O of a circle of radius 45 cm. The length of each tangent is 60 cm. Find the length of the chord AB. 3

18. In $\triangle ABC$, right-angled at B, $AB = 60$ cm and $BC = 63$ cm. A circle is inscribed in the triangle. Find the radius of the inscribed circle. 3

OR

Tangents QA and QB are drawn from an external point Q to a circle, and $QA = (23x + 4)$ cm and $QB = (15x + 20)$ cm. Find the length of QA. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. **Case study.** Two tangents PA and PB are drawn from an external point P to a circular fountain with centre O, and $\angle APB = 80^\circ$. 5
(i) Find $\angle OAP$.
(ii) Find $\angle AOB$.
(iii) Find $\angle OPA$.
(iv) Find $\angle OAB$ and $\angle PAB$.

20. Tangents PA and PB are drawn from a point P, 35 cm from the centre O of a circle of radius 21 cm. The length of each tangent is 28 cm. Find the length of the chord AB. 5

OR

In $\triangle ABC$, $\angle B = 90^\circ$, $AB = 35$ cm and $BC = 84$ cm. A circle touching all three sides touches AB at F and BC at D. Find the radius of the circle and the lengths AF and CD (in this order). 5

END OF QUESTION PAPER — **Total: 40 Marks**